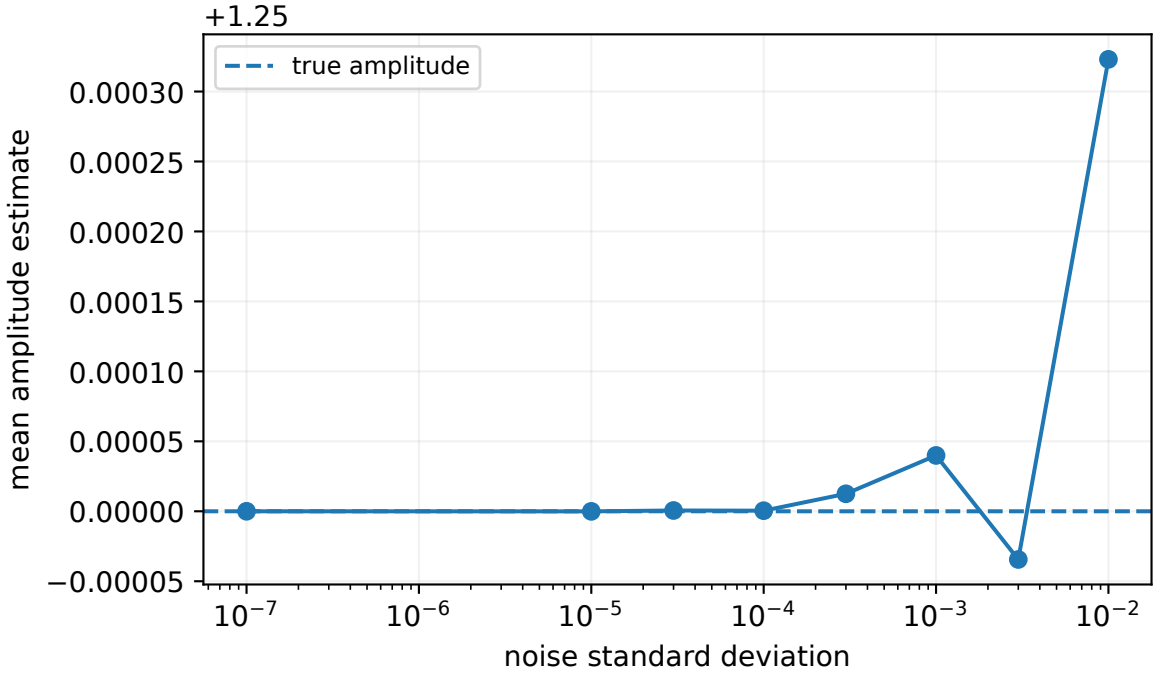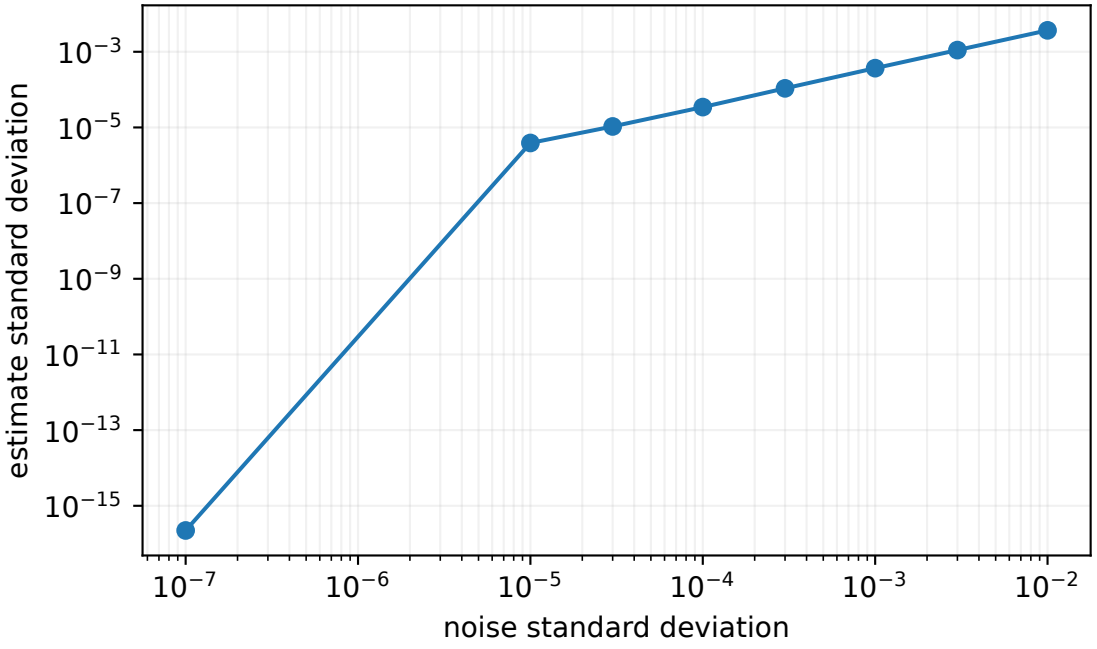The out-of-band edge functional is exact without noise and has measurable noise amplification